

CSCE 649 PHYSICALLY-BASED MODELING · MODULE 12 · LECTURE NOTES

Fluids II: Particles and Hybrids

Carrying the fluid instead of sampling it.

MODULE OUTCOMES

- Explain SPH and the role of the smoothing kernel.
- Explain why SPH struggles with incompressibility, and what PBF does about it.
- Explain PIC and FLIP and the trade between them.
- Explain MPM and what it enables.
- Choose a fluid method from the phenomenon being simulated.

1 Lagrangian fluids

	Eulerian (grid)	Lagrangian (particles)
Viewpoint	Watch fixed points in space; fluid flows past them.	Follow parcels of fluid as they move.
Advection	Dissipative — requires interpolation every step (Module 11).	Free and exact — the particles simply move, carrying their quantities with them.
Incompressibility	Easy — a global Poisson solve on a regular structure.	Hard — the constraint is global and the information is local.
Free surface	Must be tracked explicitly (level set), and loses volume.	Implicit — the surface is wherever the particles are.
Best for	Smoke, fire, confined flow.	Splashing liquid, spray, scenes with lots of free surface.

The strengths are exactly complementary, which is the entire reason hybrid methods (§2) came to dominate.

$$A(\mathbf{x}) = \sum_j m_j (A_j / \rho_j) W(\mathbf{x} - \mathbf{x}_j, h)$$

Smoothed particle hydrodynamics represents every field as a kernel-weighted sum over nearby particles. The smoothing kernel W has compact support of radius h , so each sum runs only over neighbours — found with a spatial hash, exactly as in Module 05's broad phase.

Density falls out as a special case: $\rho_i = \sum_j m_j W(\mathbf{x}_i - \mathbf{x}_j, h)$, with no separate continuity equation needed. **And derivatives move onto the kernel:** $\nabla A = \sum_j m_j (A_j / \rho_j) \nabla W$. This is the elegant part — the data is an unordered point cloud and is not differentiable, but the kernel is smooth and its gradient is known analytically.

SPH'S FUNDAMENTAL DIFFICULTY IS INCOMPRESSIBILITY

Incompressibility is a global constraint — compressing fluid here requires it to move somewhere else, instantly, however far away. **SPH has only local information:** each particle knows about its neighbours within h , and nothing else.

Classic SPH approximates it with a stiff equation of state: pressure rises steeply as density exceeds its rest value, so the fluid is strongly discouraged from compressing. It is *nearly* incompressible.

Stiff means tiny time steps by Module 02's bound — and the fluid still compresses visibly, which reads as a springy, bouncy liquid that behaves more like gelatin than water.

This is why SPH water so often looks wrong, and why a succession of methods — PCISPH, IISPH, DFSPH, and position-based fluids — exist. All of them replace the stiff-equation-of-state approximation with some way of actually imposing the constraint.

POSITION-BASED FLUIDS

PBF applies Module 04's machinery directly: treat constant density as a constraint, $C_i = \rho_i/\rho_0 - 1 = 0$, and solve it by position projection.

Unconditionally stable, for precisely the reason PBD is: positions are projected onto valid configurations rather than integrated from forces, so nothing can diverge.

Large time steps, which is what interactive applications require, and it parallelises well on a GPU.

It inherits PBD's theoretical weaknesses: the degree of incompressibility achieved depends on the iteration count and the time step, and the method is not derived from fluid mechanics. **And it is in essentially every game that has interactive fluid**, for the same reason PBD is in every game that has cloth.

2 PIC and FLIP

The hybrid idea is to use each representation where it is strong: **particles for advection, a grid for the pressure solve.**

- 1 **Transfer** particle velocities onto a MAC grid, by kernel weighting.
- 2 **Solve** for pressure and project on the grid — Module 11's machinery, in the setting where it is straightforward.
- 3 **Transfer** the resulting velocities back to the particles.
- 4 **Move** the particles. Advection is exact and costs nothing.

	PIC	FLIP
Step 3 transfers	The full grid velocity.	Only the change in grid velocity, added to the particle's existing velocity.
Dissipation	Heavy. Two interpolations per step smooth the field substantially, and a splash dies quickly.	Almost none. The particle's velocity is preserved and only corrected.
Noise	None. All particles in a cell agree.	Significant. Nothing couples particles within a cell, so their velocities drift apart — producing a jittery, sparkling liquid.

EVERYONE BLENDS THEM

Pure FLIP is too noisy and pure PIC is too damped, and both failures are immediately visible.

So blend the two results: typically 95% FLIP and 5% PIC. The small PIC component acts as a filter removing particle-scale noise, while 95% of the energy is retained. The blend costs one extra interpolation and one lerp.

That ratio is an exposed tuning parameter in every production fluid solver, and artists adjust it per shot — more PIC for a calm pool, more FLIP for a violent splash.

APIC and PolyPIC are the principled alternatives. They transfer additional information — an affine velocity field per particle — which eliminates the noise without the damping, largely removing the need for the blend parameter. They are the better answer and are increasingly standard.

3 The material point method

MPM: THE SAME HYBRID STRUCTURE, APPLIED TO EVERYTHING

MPM uses the same transfer structure as FLIP — particles to grid, solve on the grid, transfer back, move particles — but the particles carry **deformation gradients** (Module 09) rather than just velocity.

So the same solver handles solids, fluids, and everything between. A constitutive model is evaluated per particle; change it and the same code simulates elastic rubber, viscous honey, or granular sand.

Topology change is automatic. Fracture, melting, merging, and tearing require no special handling, because the grid is discarded and rebuilt every step — there is no mesh whose connectivity would have to be updated. This is the property that made it transformative for effects work.

This is the snow in Disney's Frozen, which was the result that brought MPM into graphics, and it is now the standard approach for sand, mud, snow, and viscoelastic materials.

Buys	Costs
One solver for solid, fluid, and granular materials.	Expensive — a full grid solve every step, plus particle-grid transfers in both directions.
Topology change for free.	Grid artefacts: detail is limited by cell size, and features aligned with the grid can be favoured.
Plasticity and fracture fall out of the constitutive model.	Numerical cohesion — material that has separated tends to clump back together, because the grid couples anything within a cell of anything else. The characteristic MPM artefact, and visible in a great deal of published work once you know to look.
Multi-material coupling is automatic. Sand in water needs no special treatment.	Many particles per cell are required for accuracy — typically four to eight — so memory grows quickly.

4 Choosing a method

Phenomenon	Method	Reasoning
Smoke, fire, gas.	Eulerian grid (Module 11).	Fills the domain, no free surface to track, and the grid's dissipation can be fought with vorticity confinement.
Splashing liquid, spray.	FLIP or APIC.	Exact advection and an implicit free surface from the particles, with the pressure solve on the grid. The standard choice in film.
Real-time or interactive water.	PBF.	Stable at large time steps and GPU-friendly. Accuracy is not the binding constraint.
Viscous or sticky fluid — honey, lava.	SPH or MPM.	Viscosity is natural in SPH; MPM if plasticity is also wanted.
Sand, snow, mud, granular material.	MPM.	Plasticity and topology change are exactly what it provides.

Phenomenon	Method	Reasoning
Large ocean surface.	Spectral FFT wave model.	Not a fluid simulation at all — see below.

NOT EVERYTHING THAT LOOKS LIKE FLUID IS SIMULATED

Ocean surfaces in film are almost never simulated. Tessendorf's spectral method synthesises a statistically correct ocean surface directly in the frequency domain, using measured wave spectra, and transforms it with an FFT.

It is vastly cheaper than a simulation, trivially tileable over an unlimited area, directly controllable through wind speed and fetch parameters, and it looks better than any fluid simulation at an affordable resolution — because it reproduces the correct statistics of a real ocean rather than approximating the dynamics.

A simulation is worth its cost only where the interaction matters: a ship's wake, a breaking wave, water striking an object.

So the production answer is hybrid: a spectral ocean for the bulk, a simulated region where something is happening, and the two blended and composited. Knowing when *not* to simulate is part of the skill, and it is the same judgement Module 01 described as the difference between a graphics simulation and a scientific one.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Bridson — Fluid Simulation for Computer Graphics	The PIC/FLIP chapters are the reference for §2, and the free notes cover the transfers in detail.
Müller, Charypar & Gross — Particle-Based Fluid Simulation (SPH, free)	The SPH formulation of §1 as graphics uses it.
Macklin & Müller — Position Based Fluids (2013, free)	PBF. Short, and the algorithm follows directly from Module 04.
Jiang et al. — The Material Point Method for Simulating Continuum Materials (SIGGRAPH course, free)	The definitive free introduction to §3, with code.
Tessendorf — Simulating Ocean Water (free)	The spectral ocean model of §4. Read it before simulating an ocean.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement SPH in 2D with a cubic spline kernel. Verify that the density sum reproduces the correct density for a uniform particle arrangement.
- 2 Demonstrate SPH compressibility: drop a column of fluid and measure the maximum density excess. Then increase the stiffness and report the effect on both compression and the stable time step.
- 3 Implement position-based fluids and compare stability and time step against your SPH implementation.
- 4 Implement PIC. Simulate a dam break and observe the damping — report how quickly the splash dies.
- 5 Change one line to make it FLIP. Observe the noise. Then implement the blend and find the ratio you prefer.
- 6 Plot kinetic energy over time for pure PIC, pure FLIP, and a 95/5 blend on the same scenario.
- 7 Measure volume conservation for your FLIP solver and compare against the grid-based liquid from Module 11. Report both.
- 8 Implement APIC and compare against the FLIP/PIC blend on noise and energy retention.
- 9 Implement a basic MPM solver with an elastic constitutive model, then change only the model to produce sand. Document that nothing else changed.
- 10 Demonstrate MPM numerical cohesion: separate two pieces of material and show them clumping back together. Report the separation distance at which it stops.
- 11 Implement Tessendorf's spectral ocean and compare against a FLIP simulation of the same ocean at the same cost. Be honest about which looks better.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Compare Eulerian and Lagrangian representations on four axes.
- 2 Write the SPH interpolation formula and say what the kernel does.
- 3 Why do derivatives fall on the kernel rather than the data?
- 4 Why is incompressibility fundamentally difficult for SPH, and what does classic SPH do about it?
- 5 What constraint does PBF enforce, and what does it inherit from PBD?
- 6 Give the four steps of PIC/FLIP and say what each representation contributes.
- 7 What is the single difference between PIC and FLIP, and what are the two failure modes?
- 8 Why does everyone blend them, and what does APIC change?
- 9 What do MPM particles carry, and what three things does that enable?

- 10 What is numerical cohesion?
- 11 Why are film oceans usually not simulated?